



10076 - I see a little silhouetto of a disk: Measuring protoplanetary ice and gas in absorption against a background nebula

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observations				
	1	114-426 NIRspec	NIRSpec IFU Spectroscopy	(1) 114-426

ABSTRACT

We propose deep NIRSpec IFU observations of the edge-on protoplanetary disk 114-426 seen in silhouette against the Orion Nebula. The orientation of this disk allows its ice and gas to be detected in absorption against the nebular background. NIRSpec's unique capabilities will enable an unprecedented near-IR spatial + spectral analysis of the system. Existing photometric data reveal absorption at 3 microns by water ice, but higher spectral resolution maps are needed to measure the crystallinity of this ice and to search for other important ices such as CO, CO₂, NH₃, and CH₃OH. Narrow band photometric data also suggest the disk preferentially absorbs Pa-alpha photons, indicating hot gas in the outer disk. NIRSpec will test this finding by searching for absorption in the many bright nebular gas lines. We will also search the measured absorption spectrum for lines from the disk's cooler molecular gas reservoir (e.g., CO). Finally, we will analyze the spectra of the two lobes of scattered light that flank the disks

midplane, which also show the 3 micron signature of water ice in the photometric data. 114-426 experiences an intermediate-strength external radiation field that is most common for young stars and disks in the Galaxy. These observations will thus reveal the ice and gas available for planet formation in a typical environment.

OBSERVING DESCRIPTION

We request NIRSpec IFU observations with the G235H/170LP and G395M/F290LP modes to detect ice and gas absorption features from 1.66-5.14 microns. We request dedicated leakcal exposures to improve data quality. We do not use TA, as we want the observations centered on the dark lane of the disk. The major axis of the disk will fit within the FOV of a single IFU pointing. We use a standard 4-pt dither pattern to improve spatial and spectral sampling. We use the NRSIRS2 readout mode to limit the data volume. We limit each integration to 15 groups to avoid saturation on the bright background nebular lines. The sensitivity to carry out the absorption analysis requires 2 integrations per exposure.

Proposal 10076 - Targets - I see a little silhouetto of a disk: Measuring protoplanetary ice and gas in absorption against a background ...

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	114-426	RA: 05 35 11.3300 (83.7972083d) Dec: -05 24 26.18 (-5.40727d) Equinox: J2000	Epoch of Position: 2000	
	<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>Category=Star</i> <i>Description=[Protoplanetary disks]</i> <i>Extended=YES</i>				

Proposal 10076 - Observation 1 - I see a little silhouetto of a disk: Measuring protoplanetary ice and gas in absorption against a backg...

Observation	Proposal 10076, Observation 1: 114-426 NIRspec											Fri Mar 13 18:04:32 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
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	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[Protoplanetary disks] Extended=YES											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points	
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G235H/F170LP	NRSIRS2	15	2	false	true	NONE	4	8	8870.045	265409.2
	2	G235H/F170LP	NRSIRS2	15	2	true	true	NONE	4	8	8870.045	265409.2
	3	G395H/F290LP	NRSIRS2	15	2	false	true	NONE	4	8	8870.045	265409.1
	4	G395H/F290LP	NRSIRS2	15	2	true	true	NONE	4	8	8870.045	265409.1